

HisabDo AI — Day 14

AI Budget Recommendation Explanation Layer — Taha

1. Purpose

This module converts verified backend budget recommendations into clear, natural-language explanations. The AI explanation layer does not calculate or change recommendation values. The backend recommendation engine remains the source of truth.

2. Taha — Exact Task Scope

1. Define AI explanation structure for budget recommendations.
2. Use only verified backend recommendation data.
3. Explain recommended category budgets.
4. Explain why a budget adjustment is suggested.
5. Explain overspending categories.
6. Explain potential savings opportunities.
7. Generate actionable budget-management suggestions.
8. Communicate limitations/uncertainty.
9. Ensure AI does not invent financial amounts.
10. Prepare sample AI budget recommendation responses.
11. Handle missing or insufficient data safely.

3. Correct AI Architecture

HisabDo Database → Budget Recommendation Engine → Verified Recommendation Result → AI Explanation Layer → Mobile/Web UI

The AI receives verified recommendation data from the backend. It explains that data in simple language and can produce practical observations, but it must not create new financial values.

4. AI Explanation Structure

1. **Recommendation Summary:** Short overview of the category budget recommendation.

- 2. Recommended Budget:** Explain the verified recommended amount for the category.
- 3. Why This Recommendation:** Explain the backend-provided reason, such as historical spending or budget comparison.
- 4. Overspending Explanation:** Explain categories where verified spending is above the existing budget.
- 5. Savings Opportunity:** Describe potential savings opportunities only when supported by verified data.
- 6. Actionable Suggestion:** Give a practical budget-management action based on the verified result.
- 7. Limitations / Uncertainty:** Explain limited history, missing data, or other backend-provided limitations.

5. Verified Backend Input

- Category
- HistoricalAverage
- ExistingBudget
- RecommendedBudget
- BudgetVariance
- BudgetUtilizationPercent
- RecommendationReason
- DataSufficiency
- Limitations

6. AI Explanation Rules

- Use RecommendedBudget exactly as supplied by the backend.
- Use HistoricalAverage exactly as supplied by the backend.
- Use ExistingBudget and BudgetUtilization only when available.
- Use the backend RecommendationReason to explain why an adjustment is suggested.
- Explain overspending only when the backend marks or supports it.
- Discuss savings opportunities only from verified spending/budget information.
- Keep suggestions practical and understandable.
- Communicate DataSufficiency and Limitations when present.

7. Mandatory No-Invention Rules

- Never invent a budget amount.
- Never change or round a backend recommendation into a different financial conclusion.
- Never calculate a replacement recommendation inside the LLM.
- Never invent historical spending.

- Never invent savings amounts.
- Never claim guaranteed future savings.
- Never make an affordability claim without verified supporting data.
- If a value is missing, state that it is unavailable rather than guessing.

8. Savings Opportunity Logic

AI may explain a savings opportunity when the verified backend result indicates that a category has spending above the existing budget or provides verified budget headroom. AI should not invent a savings amount unless that amount is explicitly supplied by the backend.

Example: If Food has an existing budget of PKR 20,000 and verified average spending of PKR 15,000, AI may explain that the category has PKR 5,000 of budget headroom because that variance is already provided/calculated by the backend.

9. Sample AI Responses

Recommended budget

Your recommended Food budget is PKR 15,000 per month. This recommendation is based on your verified recent spending pattern. It is an estimate based on historical behavior and may change as your spending changes.

Budget adjustment

Your current Food budget is PKR 12,000, while your verified historical spending averages PKR 15,000. The recommendation suggests reviewing the budget because recent spending is above the current limit.

Overspending

Food is currently an overspending category because your verified historical spending is above the existing budget. Reviewing this category may help you keep spending closer to your planned budget.

Savings opportunity

Your verified spending pattern shows budget headroom in Transport. You can review this category to decide whether the available budget can be retained for future needs or adjusted.

Actionable suggestion

Review your highest-spending categories first and compare them with their recommended budgets. Focus on categories marked for review rather than making changes based on assumptions.

Limited data

The budget recommendation is based on limited historical data. Treat it as an initial estimate and review it again as more transaction history becomes available.

Missing data

A complete budget recommendation cannot be explained because some verified backend data is unavailable. No missing financial amount has been estimated.

10. Recommended AI Prompt

You are the HisabDo AI budget explanation assistant. Explain only the verified budget recommendation data supplied by the backend. Do not calculate a new budget, invent financial amounts, invent categories, or change any backend value. Explain the recommendation, its verified reason, overspending or budget headroom where supplied, and practical actions. If data is missing or limited, state that clearly. Treat recommendations as estimates based on historical data, not guarantees.

11. Missing / Insufficient Data Handling

- No recommendation available → explain that verified data is insufficient.
- RecommendedBudget missing → do not create or estimate an amount.
- ExistingBudget missing → explain the recommendation without current-budget comparison.
- HistoricalAverage missing → do not claim a historical spending amount.
- Reason missing → provide only a neutral explanation; do not invent a reason.
- Limitations present → communicate them to the user.
- Limited data → label the recommendation as an estimate and avoid strong conclusions.

12. .NET Implementation Boundary

The C# service below builds a safe explanation context from verified backend values. It is suitable for passing to an LLM provider later. The LLM remains an explanation layer; all financial calculations remain in the backend recommendation engine.

13. Testing Scenarios

- Normal recommended category budget.
- Current budget below historical spending.
- Current budget above historical spending.
- Overspending category.
- Underspending/budget-headroom category.
- No existing budget.
- Limited historical data.
- Missing recommended budget.
- Missing historical average.
- Missing recommendation reason.
- Verify AI response does not invent amounts.
- Verify AI response preserves backend recommendation values.

14. Taha Acceptance Checklist

- ☐ Define AI explanation structure for budget recommendations.
- ☐ Use only verified backend recommendation data.
- ☐ Explain recommended category budgets.
- ☐ Explain why a budget adjustment is suggested.
- ☐ Explain overspending categories.
- ☐ Explain potential savings opportunities.
- ☐ Generate actionable budget-management suggestions.
- ☐ Communicate limitations/uncertainty.
- ☐ Ensure AI does not invent financial amounts.
- ☐ Prepare sample AI budget recommendation responses.
- ☐ Handle missing or insufficient data safely.

15. Final Definition

The Day 14 AI Budget Explanation Layer converts verified category budget recommendations into simple and actionable natural-language insights. It explains recommended budgets, adjustment reasons, overspending and supported savings opportunities, while clearly communicating limitations. The AI must never invent or replace financial amounts; the backend remains the source of truth.